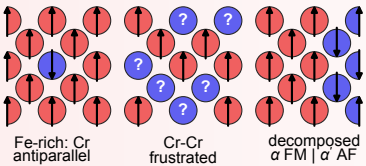
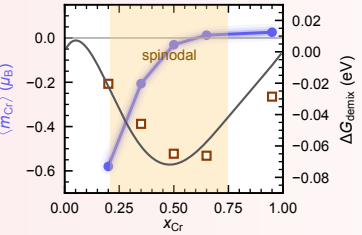
FRUSTRATE → NEST → STIFFEN → DETECT

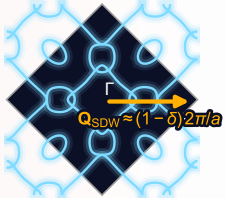
(a) atoms & moments → (b) electrons at E_F → (c) electron → lattice → (d) surface acoustic waves

schematic





Cr: jack (Γ) nests onto octahedron (H)
⇒ spin- and charge-density waves

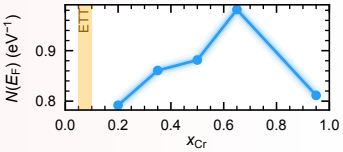


ETT (Lifshitz): adding Cr moves E_F →

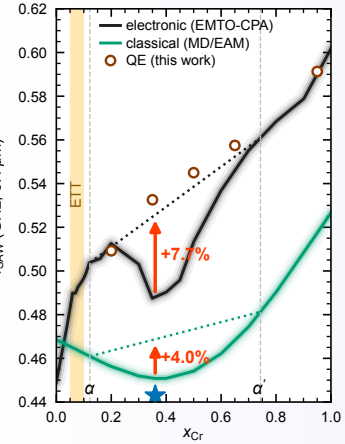


Fe ↑ sheet, 5−10% Cr: √-type cusp in B, C'

schematic



$C = C_{ion} + C_{band}$, $C_{band} \propto -\sum_k (\partial\varepsilon_k/\partial\eta)^2\, \delta(\varepsilon_k - E_F)$



10^4 grains, λ = 6.4 μm; QE at 35% Cr

